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Optimizing Non-Invasive PGT-A: A Multi-Factorial Approach for Enhanced Accuracy and Seamless Integration Into Clinical IVF

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ABSTRACT

Purpose: This study aimed to identify factors influencing the accuracy and effectiveness of non-invasive preimplantation genetic testing for aneuploidy (niPGT-A) and develop clinical integration workflows to provide alternative embryo assessment and improve pregnancy outcomes.

Methods: A two-phase study analyzed 341 blastocysts and spent culture medium (SCM) from 90 IVF patients. Phase 1 assessed niPGT-A accuracy by comparing SCM/trophectoderm (TE) with whole blastocysts (100 frozen embryos). Phase 2 examined concordance, assisted hatching, and culture-duration effects, focusing on outcomes after frozen embryo transfer using 241 fresh embryos. Nineteen euploid-TE/euploid-SCM and 14 euploid-TE/aneuploid-SCM blastocysts were transferred, and a meta-analysis was conducted using data from 163 euploid embryos across four studies.

Results: The optimized niPGT-A workflow achieved a superior PPV of 92.1% and accuracy of 91.3%, outperforming conventional PGT-A. Ploidy and sex concordances were 76.5% and 98.9% among 183 fresh embryos. Assisted hatching improved amplification rates, while culture duration/ploidy did not impact concordance. Meta-analysis showed SCM-guided embryo transfer significantly improved pregnancy outcomes, compared to TE biopsy.

Conclusion: This optimized niPGT-A procedure provides a highly accurate chromosomal assessment, seamlessly integrates with IVF workflows. It supports informed embryo transfer decisions, improves outcomes, reduces embryo damage risk, and limits invasive procedures, enhancing clinical management in reproductive medicine.

1 | Introduction

Preimplantation genetic testing for aneuploidy (PGT-A) is a crucial tool for evaluating chromosomal abnormalities in embryos before transfer, which has notably improved live birth rates, particularly in women of advanced maternal age [1–3]. Despite a clinical pregnancy rate of 46% [4] and a live birth rate of 40% [5] in older women using trophectoderm (TE) biopsy-based

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PGT-A, significant concerns persist regarding the method's accuracy, invasiveness, and safety [6–9]. TE biopsies are prone to producing false positive and negative results due to sampling limitations, as highlighted by a recent study using 390 blastocysts, and a multifocal biopsy approach (four TE and one inner cell mass, ICM) has revealed 96.1% and 32.1% uniform concordances for whole chromosome and segmental aneuploidies across multifocal analysis, respective [10]. Additionally, poor-quality embryos are particularly susceptible to damage during TE biopsy, which exacerbates concerns about embryo quality and viability [11]. These limitations underscore the pressing need for a reliable, non-invasive alternative.

During the early development of an embryo, embryonic cells usually undergo apoptosis to maintain correct differentiation or to exclude chromosome and nuclear anomalies [12]. Non-invasive preimplantation genetic testing for aneuploidy (niPGT-A) [13–16] emerges as a promising solution, leveraging the detection of embryonic cfDNA in blastocoel fluid (BF) [17] and spent culture medium (SCM) [18]. This approach offers a less invasive means of assessing the ploidy state, with the added benefit of preserving embryonic integrity [19]. Given the higher apoptotic frequency within the ICM [20, 21], niPGT-A may more accurately reflect the embryo's true genetic status compared to TE biopsies, and provide information on cell-free DNA fragmentation index (cfDFI) [22]. Some laboratories have reported high success rates in cfDNA amplification and interpretation [16, 23, 24], alongside commendable concordance with PGT-A results [14, 24, 25]. However, variable outcomes across IVF centers have impeded its routine clinical implementation, often attributed to differences in techniques such as assisted hatching (AH) [26], incubation times (24–72 h) [27], and whole genome amplification (WGA) reagents [28].

Addressing these inconsistencies is critical to maximizing the potential benefits of niPGT-A, particularly its less technically demanding nature, potential for increased accessibility, and the promise of improved overall accuracy compared to traditional PGT-A. Therefore, this study aims to systematically evaluate critical procedural factors, including WGA reagents, AH protocols, and culture duration, to develop a refined clinical niPGT-A protocol. By using whole blastocysts (WBs) or the ICM as benchmarks, we aim to optimize niPGT-A methods to enhance accuracy and clinical outcomes, particularly for frozen–thawed embryo transfers. Clarifying and standardizing these procedural elements will not only address current gaps in the literature but also pave the way for the broader adoption of niPGT-A in routine IVF practice, ultimately providing patients with a safer and more efficacious option for genetic screening and improved clinical outcomes in routine IVF clinics for frozen–thawed embryo transfers.

2 | Materials and Methods

2.1 | Study Subjects and Ethics Statement

Ninety patients receiving infertility treatment at the An-An Women and Children Clinic in Tainan City, Taiwan, participated in this study from June 2021 to August 2022. This study was approved by the Institutional Review Boards (IRB, A-ER-110-137) from the National Cheng Kung University Hospital, Taiwan. Written informed consent was obtained from all participants, and the study adhered to relevant guidelines and regulations. Participant demographic information is detailed in Table 1.

2.2 | Clinical Procedures, Sample Collection, and Study Design

The sample collection process was conducted with utmost care and precision. One hundred donated frozen embryos,

TABLE 1 | The demographic information of participants.

Demographic information	Donated frozen embryos		Fresh embryos	
Processing	WB/AH	ICM/no-AH	AH	No-AH
Patient				
Patient number	23	6	60	8
Age (mean ± SD) (range)	38.4 ± 4.8 (30–45)	36.2 ± 3.8 (31–42)	38.0 ± 3.8 (27–44)	34.9 ± 4.7 (28–43)
Embryos	80	20	208	33
Sampling				
WB/ICM	80	20	—	—
Biopsied TE	80	20	208	33
SCM	80	20	208	33
Culture time (h)				
1-day (17–25)	80	20	12	0
2-day (42–52)	0	0	138	33
3-day (65–91)	0	0	58	0

Abbreviations: AH, assisted hatching; ICM, inner cell mass; SCM, spent culture media; TE, trophectoderm; WB, whole blastocyst.

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regardless of their morphological grade, fertilized through either conventional IVF ($N=8$) or ICSI ($N=92$) and vitrified (VT601; Kitazato, Shizuoka, Japan), were thawed and washed extensively with blastocyst culture medium (CooperSurgical, Ballerup, Denmark). The embryos were cultured in 20 μL of medium for 24 h. Among them, 80 embryos underwent AH before cryopreservation and were subjected to TE biopsy alongside 15 μL SCM collection in RNase- and DNase-free PCR tubes for niPGT-A analysis. For the remaining 20 embryos, 15 μL of culture medium was collected after 24 h without AH, followed by TE biopsies and ICM isolation.

To establish a fresh culture niPGT-A protocol, 241 embryos from patients undergoing an IVF cycle with ICSI were cultured using a standard two-step embryo culture system in clinical settings. Fertilized embryos were pooled and cultured in media until day 3 or the embryo reached the 8-cell stage, then transferred to a second stage of individual culture in 20 μL media droplets until the embryo reached the blastocyst stage (i.e., day 5 or 6). To study the AH effect, 208 embryos underwent AH at the morula stage, while 33 did not, with SCM and TE biopsy performed at the end of the culture. During embryo culture, a droplet of culture media alone, without any embryo, serves as a blank control.

Blastocysts were graded according to the Gardner and Schoolcraft system based on the degree of blastocyst expansion and the morphological appearance of the ICM and TE cells. According to a previous publication [29], blastocyst morphologies were assessed and categorized into four groups (excellent, good, average, and poor quality) to analyze the relationship between grade and ploidy state.

2.3 | Whole Genome Amplification, Library Preparation, and Next-Generation Sequencing

TE biopsies and WB/ICM samples underwent a TFDA-approved PGT-A workflow using the PicoPLEX Gold Single Cell DNA-Seq kit (Takara Bio, Mountain View, CA, USA). SCM samples were processed with WGA using either PicoPLEX Gold ($N=100$), PG-Seq Rapid Non-Invasive PGT kit (PerkinElmer, Waltham, MA, USA, $N=40$), or NICSInst (Yikon Genomics, Shanghai, China, $N=40$). WGA and NGS library preparations followed manufacturer protocols, with library quality assessment conducted via microfluidic capillary electrophoresis on LabChip GX/GX II (Caliper Life Sciences, Hopkinton, MA, USA). The PicoPLEX Gold kit generated WGA libraries with a 400–900 bp size range, while NICSInst and PG-Seq libraries ranged from 400 to 600 and 300 to 400 bp, respectively. Blank medium controls showed no amplification, and appropriately sized libraries proceeded to qPCR measurement using KAPA Library Quantification kits (Roche, Basel, Switzerland) on a StepOnePlus Real-time PCR System (Thermo Fisher Scientific, Waltham, MA, USA). Libraries with concentrations below 1 nM were deemed amplification failures; successful libraries were sequenced using a NextSeq550 Illumina Sequencer (Illumina Inc., San Diego, CA, USA) with approximately 1–2 million single-end (1×150 bp) reads per sample.

2.4 | Bioinformatics Analysis

High-quality reads were aligned to the UCSC human reference genome (hg19) and processed through bioinformatics pipelines to determine chromosomal ploidy. NICSInst- and PG-Seq-amplified embryos were analyzed with ChromGo (Yikon Genomics, Shanghai, China) and PG-Find software (PerkinElmer, Waltham, MA, USA), respectively. Meanwhile, PicoPLEX Gold samples were evaluated using an in-house bioinformatics pipeline [30]. ChromGo utilized whole-arm resolution with a 30%–70% mosaic threshold, while PG-Find applied a 5 Mb CNV resolution, defining aneuploidy as CN > 2.7 (trisomy) or < 1.3 (monosomy).

The in-house pipeline divided the reference genome into 1 Mb bins, normalizing reads to a total coverage of 1; regions with no reads received a placeholder value of 0.0000001. The ratio of regional coverages was converted to the chromosomal copy number. In euploid embryos, the copy number was 2; aneuploidy was identified when more than 10 Mb showed an 80% copy gain or loss. Mosaicism was initially defined as a 40%–80% deviation from the normal CN of 2 but was refined to 40%–70% to improve niPGT-A accuracy. Segmental aneuploidy was reported when chromosomal changes exceeded 10 Mb, with chaotic CN identified when over five chromosomes exhibited trisomy or monosomy.

2.5 | Female Cell Contamination

niPGT-A aims to analyze the genetic material originating from the embryo. However, female cell contamination can occur. Therefore, it is important to estimate the proportion of female cell contamination to ensure the accuracy and reliability of the test results. Estimating female cell contamination in a predominantly male cell sample, or vice versa, can be achieved by analyzing the X:Y chromosome ratio. Since female cells (XX) contribute two X chromosomes while male cells (XY) contribute one X and one Y, deviations from a 1:1 X:Y ratio indicate contamination. We can estimate the proportion of female cells present by measuring the deviation from the expected X:Y ratio. This calculation assumes that the primary sources of X and Y chromosomes are the respective cell types and that the measurement of the X:Y ratio is accurate, allowing for a reasonable approximation of the contamination percentage.

Given the circumstance, a mixture of female (XX) and male (XY) cells where F represents the proportion of female cells and M represents the proportion of male cells ($F+M=1$), $2F+M$ represents the total X chromosomes, and total Y chromosomes by M. Given a measured X:Y ratio, we can then solve for F. For example, with an X:Y ratio of 1.8:1, the equation $(2F+M)/M=1.8$ simplifies to $F=0.4$ M. Substituting $F=1-M$, we find $M\approx0.714$ and $F\approx0.286$, suggesting approximately 28.6% female cell contamination.

2.6 | Pregnancy Outcomes

The primary outcome was the live birth rate, calculated as the number of participants with at least one live birth after

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24 weeks of gestation divided by the total number of cycles. Secondary outcomes included the implantation rate, clinical pregnancy rate, ongoing pregnancy rate, and miscarriage rate. The implantation rate was defined as the number of embryos with a positive serum β-hCG level (≥ 30 mIU/mL) generally considered positive for pregnancy, divided by the number of embryos transferred. The clinical pregnancy rate was determined by the number of visible gestational sacs on ultrasound divided by the number of embryos transferred. In this study, miscarriage was defined as spontaneous pregnancy loss before 20 weeks of gestation. Ongoing pregnancy was defined as confirmation of a viable pregnancy reaching at least 20 weeks of gestation. Deliveries occurring between 20 and 23 weeks were categorized as live births if any signs of life (e.g., heartbeat, breathing) were present at delivery, regardless of subsequent viability. In the absence of such signs at delivery, the event was classified as stillbirth.

2.7 | Statistical Analysis

The performance of niPGT-A and PGT-A was evaluated using ICM or WB as references to determine true positive (TP), false negative (FN), true negative (TN), and false positive (FP) rates. Clinical statistics were calculated for positive predictive value (PPV), negative predictive value (NPV), sensitivity, specificity, accuracy, and concordance rate between PGT-A and niPGT-A from fresh embryos. Data analysis involved paired/unpaired Student's *t*-test, one-way ANOVA with Tukey's multiple tests, and chi-squared tests using GraphPad Prism 6.0 (GraphPad Software, La Jolla, CA, USA).

Four datasets from three previous studies that applied similar classification and compared various pregnancy outcomes between TE-called and SCM-called ploidy results [25, 28, 31] were used to conduct a meta-analysis following the procedures outlined by Simopoulou et al. [32] A chi-squared test for heterogeneity was conducted, with *p*-values calculated. A fixed-effects model was used when no statistically significant heterogeneity was present ($I^2 < 50\%$), and a random-effects meta-analysis was applied when substantial heterogeneity was observed ($I^2 > 50\%$). Results were expressed as risk ratios (RR) with 95% confidence intervals (CI) and graphically summarized in a forest plot for various outcomes. All data are reported as mean ± SE, with statistical significance denoted by $p < 0.05$.

3 | Results

3.1 | Establishment of a niPGT-A Protocol

To establish a robust standard niPGT-A protocol in a clinical setting, we included 80 donated embryos from 23 patients aged 30–45 (mean age = 38.4 ± 4.8 years; Table 1). Biopsied TE cells, SCM, and WB were collected from identical embryos and subjected to the WGA procedure. WB ($N = 80$) and TE biopsy ($N = 80$) samples were prepared using an in-house, TFDA-approved PGT-A pipeline with the PicoPLEX Gold kit. SCM samples were amplified and constructed into NGS libraries using three different kits—PicoPLEX Gold ($N = 80$), PG-Seq ($N = 40$), and NICSInst kit ($N = 40$)—to evaluate WGA and library preparation performance (Figure 1A). Except for three TE biopsies (3.8%) that could not be interpreted due to noisy signals, all WB and SCM samples were successfully amplified and sequenced with approximately 1–2 million, 1 × 150 bp single-end reads.

We compared WGA performance among SCM samples amplified with different kits by evaluating library concentration and mapping rate. We found that PG-Seq showed the highest average library concentration (232.1 ± 329.2 nM, ranging 0.22–1143.6 nM, $p < 0.05$, Figure 1B) compared to NICSInst (156.8 ± 139.3 nM, ranging 6.7–524.9 nM) and PicoPLEX Gold (133.5 ± 91.6 nM, ranging 5.8–324.0 nM). Nevertheless, the large variation in concentrations with PG-Seq suggests potential instability in the WGA process. PicoPLEX Gold demonstrated a superior mapping rate (98.7% ± 1.3%) compared to NICSInst (97.5% ± 1.5%, $p < 0.001$, Figure 1C, left) and PG-Seq (95.6% ± 4.3%, $p < 0.0001$, Figure 1C, right). Despite variations in WGA performance, all three kits produced libraries suitable for high-throughput sequencing.

Next, the chromosome copy number results from different software were compared. Initially, ploidy calls showed high concordance, with rates of 82.5% (33/40) between PicoPLEX Gold and NICSInst, and 80.0% (32/40) between PicoPLEX Gold and PG-Seq (Figure 1D,E, left). We then used CN results from the WB as a gold standard to assess the performance of the niPGT-A among three kits (Figure 1D,E, right). Using a mosaicism cutoff of 40%–80% (Table 2), PicoPLEX Gold demonstrated the highest sensitivity at 96.4%, while PG-Seq achieved the highest PPV value at 89.3%. Both PicoPLEX Gold and PG-Seq achieved an accuracy rate of 82.5%, surpassing NICSInst's accuracy rate of 75%. Given that sensitivity is crucial in developing a screening test, PicoPLEX Gold was selected to advance the niPGT-A clinical protocol. Remarkably, the performance of niPGT-A using PicoPLEX Gold was comparable to that of PGT-A, without needing additional optimization (Table 2).

3.2 | Optimization of Clinical niPGT-A Protocol

Supports the previous notion that embryos may undergo a "self-correction" process to expel DNA from apoptotic aneuploid cells [21, 33], which can result in the accumulation of abnormal cfDNA in SCM and potentially lead to false positive mosaicism calls in niPGT-A. We observed that over half of the embryos (43 out of 80, or 53.8%) displayed mosaic chromosomes when using a conventional mosaicism cutoff of 40% in SCM samples.

To optimize performance, we tested various thresholds for accurate mosaicism identification in paired WB and SCM embryos (Figure 2A–C). The 40% cutoff had a high false positive rate (FPR, 72.5%) with no false negatives. Increasing to 70% reduced false positives to 2% but raised false negatives to 3.8% (Figure 2D). The 60% cutoff yielded the best diagnostic accuracy,

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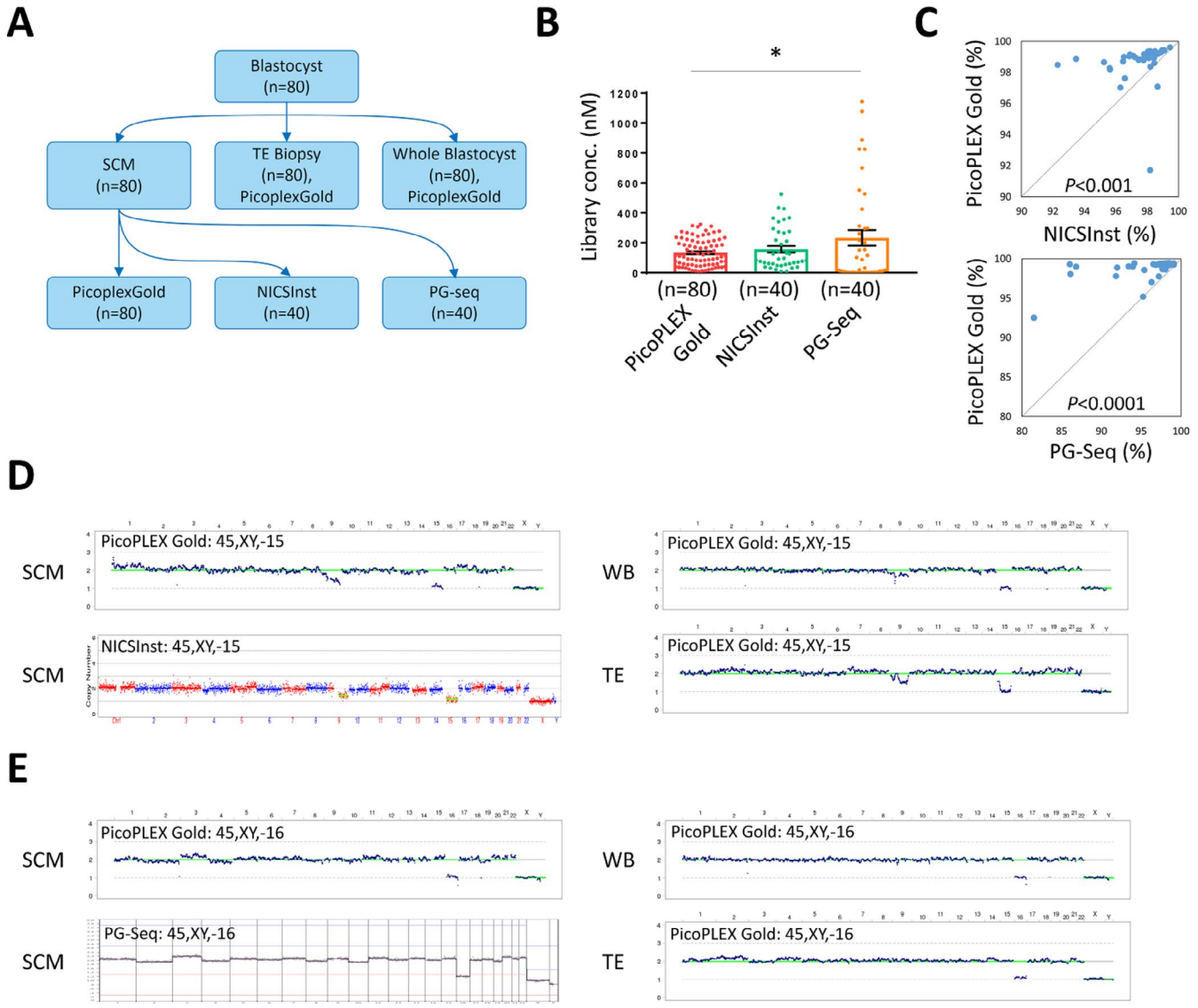


FIGURE 1 | Study design and workflows for establishing the niPGT-A standard protocol. (A) A flow chart detailing the sample processing steps for niPGT-A and PGT-A using specific commercial WGA kits for frozen embryos. (B) A comparison of library concentrations of SCM samples processed with various WGA kits, showing significant differences ($p < 0.05$) calculated using one-way ANOVA. (C) A comparison of the mapping rate of sequencing reads from niPGT-A libraries using different kits, with significant differences determined by a paired *t*-test. (D, E) Representative data illustrating concordant ploidy calls among PicoPLEX Gold, NICSInst, and PG-Seq, as indicated above each plot. The statistical significance levels are denoted as follows: $*p < 0.05$.

TABLE 2 | Comparison of performance between PGT-A and niPGT-A with different commercial kits by using the whole blastocyst as the gold standard.

Assay (WGA kit)	PGT-A (PicoPLEX Gold)	niPGT-A (PicoPLEX Gold)	niPGT-A (NICSInst)	niPGT-A (PG-Seq)
Sample number	80	80	40	40
Call rate (%)	96.2	100	100	100
PPV (%)	78.5	81.5/92.1	75.9/88.0	89.3/100
NPV (%)	83.3	86.7/88.2	72.7/66.7	66.7/58.3
Sensitivity (%)	96.2	96.4/96.7	88.0/81.5	86.2/84.8
Specificity (%)	41.7	52.0/75.0	53.3/76.9	72.7/100
Accuracy (%)	79.2	82.5/91.3	75.0/80.0	82.5/87.5

Note: The mosaicism thresholds for PGT-A and all niPGT-A data in the left column range from 40% to 80%. In the right column, the mosaicism threshold for niPGT-A data ranges from 60% to 80%.

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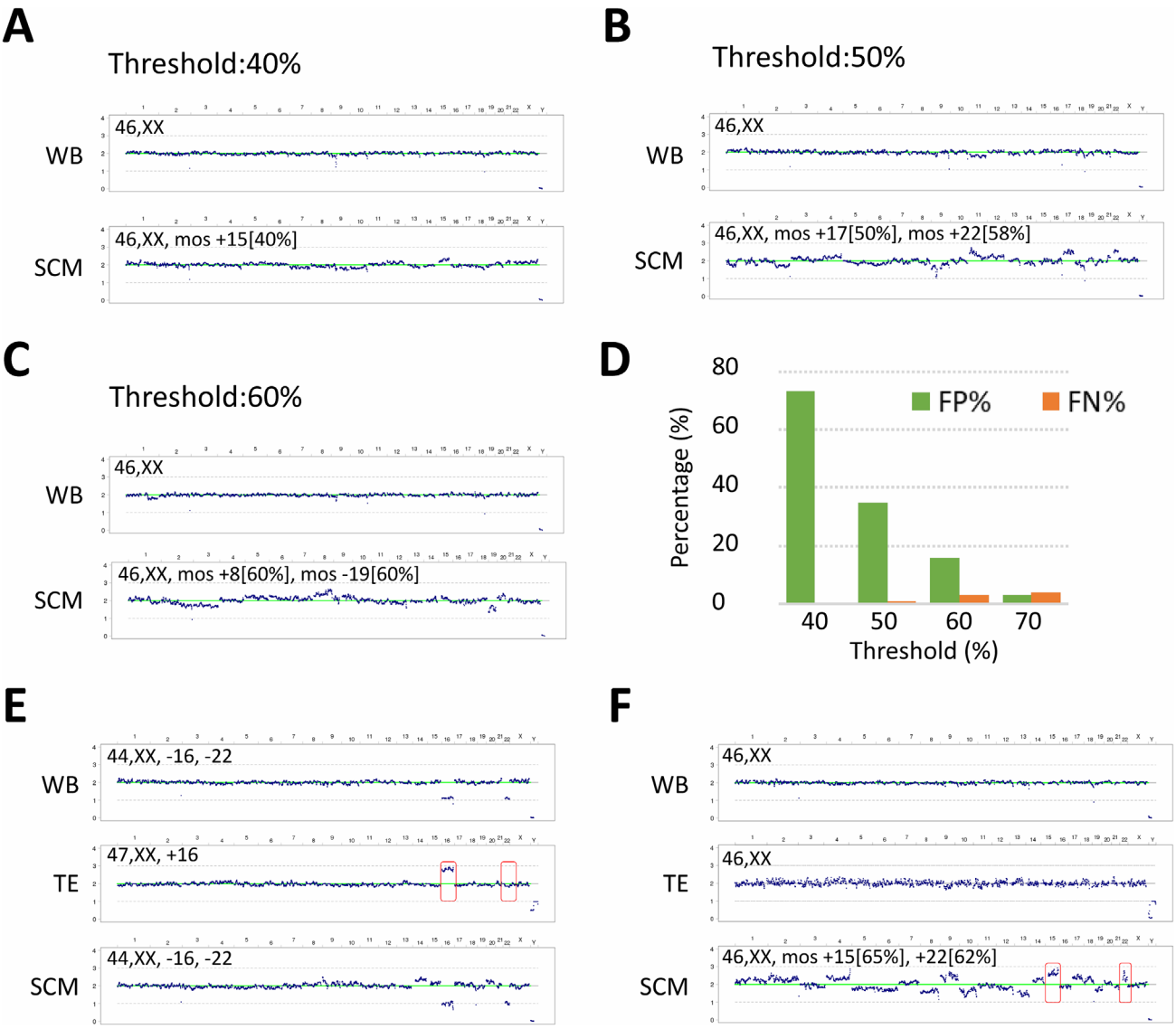


FIGURE 2 | Optimization of mosaicism threshold for niPGT-A. (A–C) Representative data showing WB and mosaic SCM samples at mosaicism thresholds of 40% (A), 50% (B), and 60% (C), with ploidy calls indicated above each plot. (D) A range of mosaicism thresholds with their corresponding false-positive and false-negative rates. (E, F) Data illustrate the discordance between TE biopsy and SCM, comparing ploidy calls to WB, SCM, and TE.

with a FPR of 16.3% and a false negative rate of 1.3%, classifying embryos as 76.3% aneuploid, 2.5% mosaic, and 21.2% euploid. This adjustment enhanced the PPV from 81.5% to 92.1% and increased specificity from 52.0% to 75.0%, resulting in an overall accuracy of 91.3%, which surpassed PGT-A's accuracy of 79.2% ($p = 0.033$, Table 2). While similar performance improvements were observed with NICSInst and PG-Seq using the 60% threshold, PicoPLEX Gold demonstrated superior accuracy at 91.3%, compared to 80.5% for NICSInst and 87.5% for PG-Seq.

The concordance assessment achieved a 77.9% (60/77) ploidy concordance rate between niPGT-A and PGT-A. Among 17 discordant samples with matched WB results, SCM correctly identified ploidy in 13 cases (76.5%, Figure 2E), while TE biopsies were accurate in only 4 cases (23.5%, Figure 2F). These findings suggest that SCM outperformed TE biopsies in accurately representing embryos' chromosomal profiles, indicating the potential of niPGT-A for routine IVF applications.

3.3 | Assessment of Library Concentration With Extended Culturing Time and Embryo Ploidy

To evaluate the feasibility of the established niPGT-A workflow for fresh embryos undergoing IVF treatment, we enrolled 208 embryos from 60 patients aged 27–44 (mean age = 38.0 ± 3.8 years). After changing medium and switching to individual culture at D3 or D4, niPGT-A was conducted on SCM samples collected from embryos cultured for 1 day (17–25 h, $N = 12$), 2 days (42–52 h, $N = 138$), or 3 days (65–91 h, $N = 58$), with PGT-A conducted on all matched TE biopsies (Figure 3A). As anticipated, extended culture durations significantly increased library concentrations (Figure 3B, $p < 0.01$), with mean concentrations of 25.5 ± 24.0 nM, 49.6 ± 67.0 nM, and 87.1 ± 78.5 nM for 1-, 2-, and 3-day cultured SCM, respectively. Although all TE biopsies were amplified properly, low library concentrations prevented downstream niPGT-A procedures in 8.3% (1/12, 1-day), 11.6% (16/138, 2-day), and 1.7% (1/58, 3-day) cultures.

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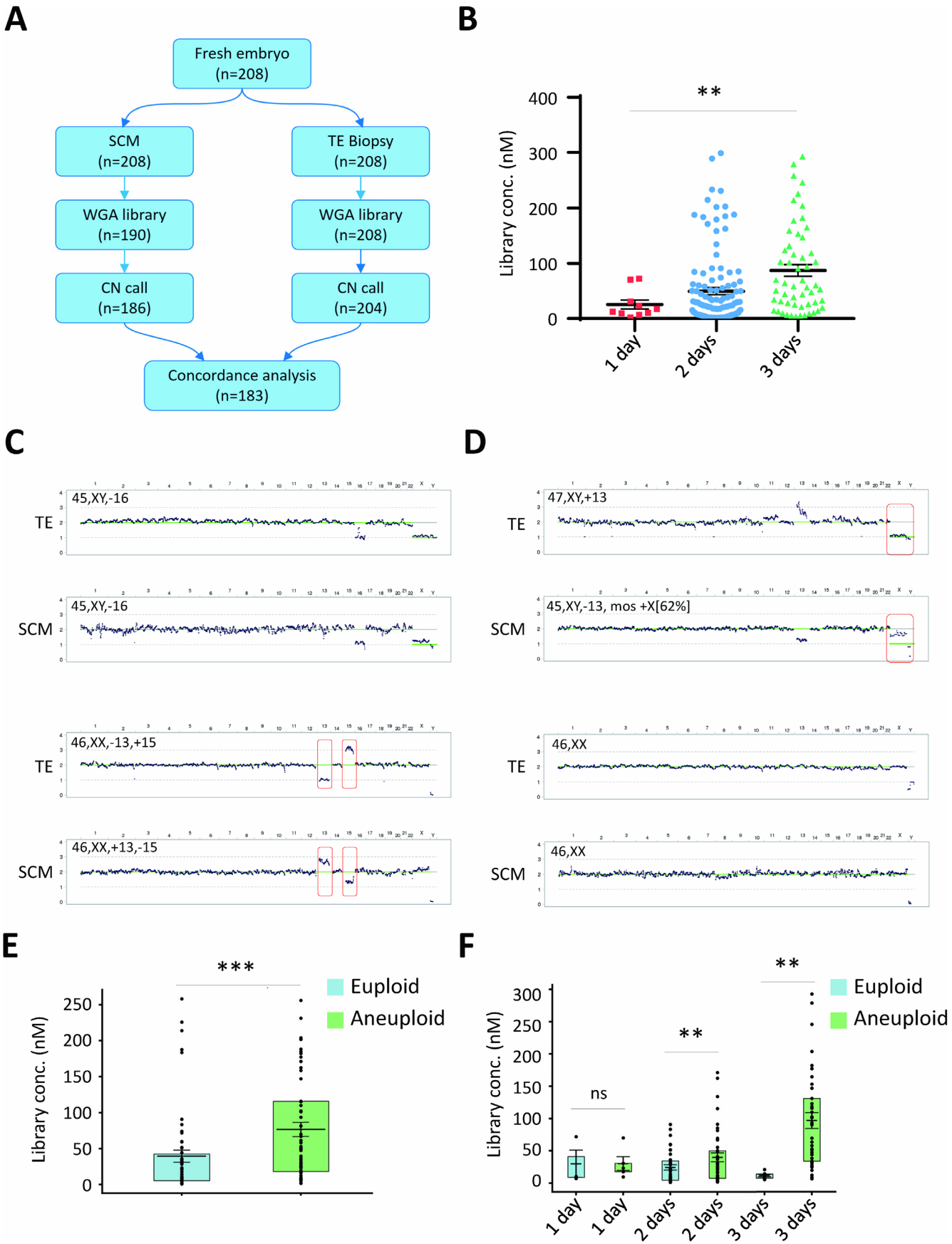


FIGURE 3 | Legend on next page.

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FIGURE 3 | The clinical application of niPGT-A in fresh embryos. (A) Flow chart detailing the sample processing for niPGT-A and PGT-A in fresh embryos, with sample numbers indicated. (B) Comparison of library concentrations in SCM samples cultured for 1–3 days, showing significant differences ($p < 0.01$) calculated by one-way ANOVA. (C) Representative data illustrating ploidy concordance (upper) and discordance (lower) between TE and SCM samples. (D) Representative data showing bioinformatics pipeline detects ~33% maternal cell contamination in male SCM sample (upper). The sex chromosome calls are the same in paired SCM and TE-biopsy female samples (bottom). (E) Box plot with a 95% confidence interval presenting library concentrations of euploid and aneuploid embryos, with a significant increase in aneuploid embryos. (F) Box plot with a 95% confidence interval demonstrating the impact of culture duration on library concentration, stratified by embryo condition, with significant increases in aneuploid embryos after 2- and 3-day cultures. The statistical significance levels are denoted as follows: ns: Not significant, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

We analyzed data from 190 SCM and 208 TE-biopsy samples using in-house bioinformatics pipelines, achieving call rates of 97.9% and 98.0%, respectively [30]. The ploidy concordance between PGT-A and niPGT-A was 76.5% in 183 paired SCM and TE-biopsy samples (Figure 3C), with similar concordance rates across 1-day (63.6%), 2-day (73.5%), and 3-day (83.6%) culture times ($p = 0.214$). Notably, the sex concordance rate reaches 98.9% among 183 fresh embryos, and only two paired TE-biopsy and SCM samples exhibited male-to-female sex discordance (2 out of 183, 1.1%). In addition, our bioinformatics pipeline detects various degrees of maternal cell contaminations, ranging from 7.6% to 45.9%, among 15 paired SCM and TE-biopsy samples that showed male karyotypes (Figure 3D). As indicated in Figure 3D, female cell contamination may affect gender recognition, but with less effect on aneuploidy calling. Furthermore, we found that library concentrations, an approximate measure of cfDNA in SCM, were significantly higher in aneuploid embryos (mean = 83.2 ± 80.7 nM) compared to euploid embryos (mean = 39.5 ± 60.6 nM, $p < 0.001$, Figure 3E); this supports the notion that most cfDNA originates from apoptotic aneuploid cells. The significant difference persisted even when stratified by culture duration, particularly in the 2-day and 3-day cultured groups ($p < 0.01$, Figure 3F). However, neither the duration of embryo culture nor ploidy affected the call rate of the niPGT-A procedure. These findings support the potential of the established niPGT-A pipeline for routine IVF applications.

3.4 | Evaluation of Embryo Grading on niPGT-A Ploidy Assessment

Next, we evaluated 181 embryos with an expansion score of 3 or higher, as determined by the Gardner embryo grading system, to investigate whether embryo grading is associated with niPGT-A results. Comparing ploidy calls across grade groups, we found no aneuploidy rate differences in 15 B3 (66.7%), 17 B4 (76.5%), and 149 B5 (67.1%) embryos ($p = 0.731$). Using Capalbo et al. [29] criteria, embryos were categorized as excellent (AA, $N = 86$), good (AB and BA, $N = 45$), average (BB, AC, and CA, $N = 47$), and below average (BC and CB, $N = 3$). The percentage of aneuploidy was 54.7% for excellent, 75.6% for good, 83% for average, and 100% for below average. Although aneuploidy increased with lower morphological grades, the differences were not statistically significant. Notably, over half of the excellent-grade embryos were aneuploid, and more than 15% of the average-grade embryos were euploid, suggesting that morphological grading alone is not a reliable predictor of ploidy. Moreover, niPGT-A offers an alternative ploidy assessment, especially for lower-grade embryos where TE biopsy poses risks.

3.5 | Increasing the Interpretable Rate of niPGT-A With the Assisted Hatching

To evaluate the necessity of AH in the niPGT-A pipeline, we collected SCM samples from 20 donated blastocysts and 33 fresh embryos without prior AH (no-AH group) (Figure 4A). ICM and TE cells were isolated in parallel from the matched embryos for PGT-A processing. We first examined library concentrations and found significantly lower levels in the no-AH group compared to the AH group (Figure 4B). For donated blastocysts, the no-AH mean was 7.8 ± 7.6 nM versus 133.5 ± 91.6 nM in the AH group ($p < 0.001$). For fresh embryos, the no-AH mean was 4.8 ± 7.1 nM vs. 60.7 ± 73.1 nM in the AH group ($p < 0.001$). Consequently, amplification failed in 15% of the no-AH group, significantly higher than the 0% failure rate in the AH group for donated embryos ($p < 0.001$). The failure rate for fresh embryos was also higher in the no-AH group at 27.3% compared to 8.6% in the AH group ($p < 0.001$). These findings indicate that AH enhances library concentration, likely by increasing cfDNA availability, which is crucial for the success of the niPGT-A procedure.

We assessed ploidy call accuracy and found that the no-AH group achieved correct ploidy calls in 80% (12/15) of cases, while the AH group had a higher accuracy rate of 91.3% (73/80), though this difference was not statistically significant ($p = 0.193$). Similarly, no difference was found in 66.7% (16/24) for the no-AH group and 76.5% (140/183) for the AH group ($p = 0.293$) in fresh embryos. These results show that while AH improves the interpretable rate of niPGT-A, overall accuracy is not influenced.

3.6 | Clinical Outcomes Evaluations

Thirty-three blastocysts were transferred to 24 patients over 28 cycles, including 23 single embryo transfer (SET) and five double embryo transfer (DET), followed by the final delivery outcome (Table 3). Among these, 16 blastocysts were identified as euploid by both TE biopsy and SCM samples (euploidy-TE/euploidy-SCM), while 12 were euploid in TE biopsy but aneuploid in SCM (euploidy-TE/aneuploidy-SCM). In the euploidy-TE/euploidy-SCM group, 11 blastocysts led to clinical pregnancies, with 10 resulting in live births. In contrast, the euploidy-TE/aneuploidy-SCM group yielded five clinical pregnancies and four live births (Table 3). Consistent with previous findings in this study, which showed that niPGT-A exhibited higher accuracy (91.3%) compared to PGT-A (79.2%), we observed greater live birth rates in the euploidy-TE/euploidy-SCM group (57.9%) compared to the other group (28.6%).

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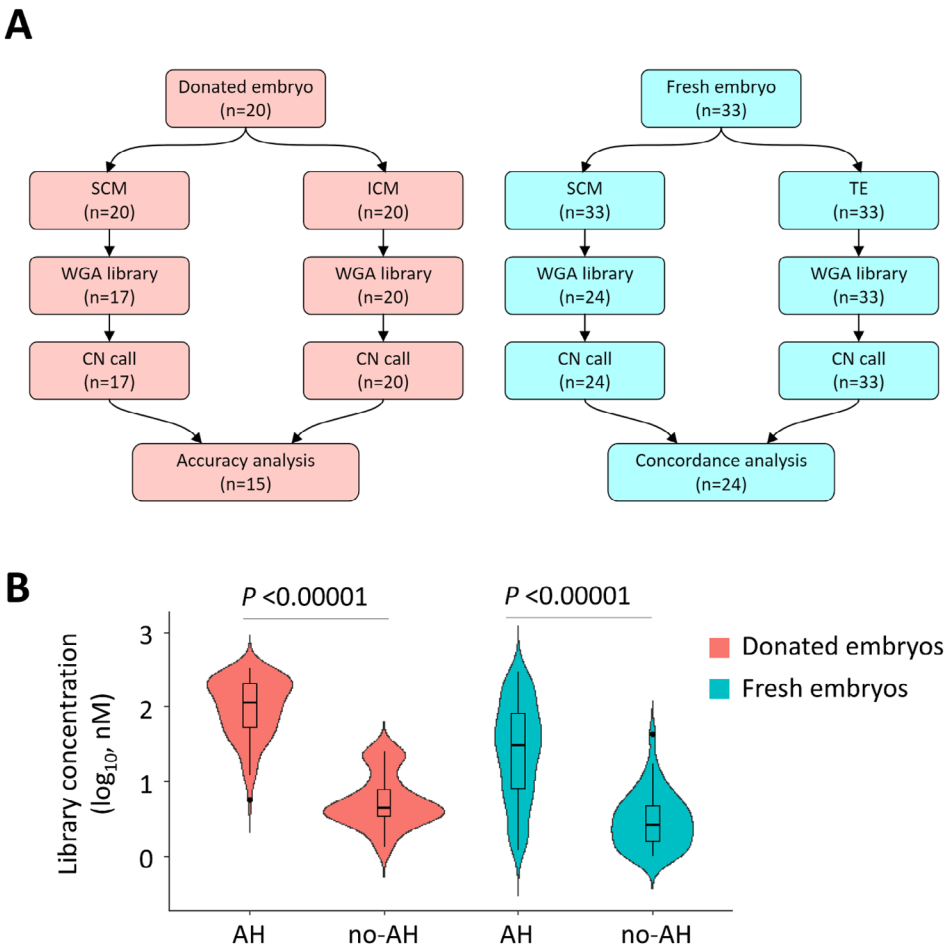


FIGURE 4 | Impact of assisted hatching (AH) on niPGT-A interpretability. (A) Flowcharts describe the processing of niPGT-A and PGT-A samples in donated and fresh embryos, emphasizing the collection of SCM without AH, followed by ICM isolation or TE biopsy. The number of samples (*n*) at each stage is indicated. (B) Violin plot showing library concentrations of embryos with and without AH, with a 95% confidence interval. *p*-values from a two-tailed unpaired *t*-test indicate a significant increase in library concentration in the AH group compared to the no-AH group for both donated and fresh embryos.

A meta-analysis was conducted to evaluate the joint clinical outcomes and improve the quantitative precision. This analysis incorporated four datasets from three previous studies that compared various pregnancy outcomes between TE-called and SCM-called ploidy results [25, 28, 31]. Among 163 single embryo transfer events, 98 embryos were classified as euploidy-TE/euploidy-SCM, and 65 events were classified as euploidy-TE/aneuploidy-SCM. The calculated heterogeneity among the four datasets was 0% ($I^2=0\%$) for the different outcomes, allowing for the application of fixed-effects models. The forest plots illustrate results from pooled datasets, showing significantly higher clinical ($p=0.039$, Figure 5a) and ongoing pregnancies ($p=0.011$, Figure 5b) for euploidy-TE/euploidy-SCM embryos compared to euploidy-TE/aneuploidy-SCM embryos, with no miscarriages observed. These findings highlight the enhanced accuracy of embryo transfer decisions guided by niPGT-A results, reinforcing the protocol's integration into routine IVF clinics for better patient outcomes.

4 | Discussion

We developed a clinically validated niPGT-A protocol that enhances the accuracy and accessibility of genetic testing in reproductive medicine. By optimizing whole genome amplification methods and incorporating AH, our optimal embryo assessment workflow not only maintains consistency with existing laboratory practices but also enhances embryo assessment without the technological demands of invasive biopsies, echoing the call for safer alternatives. We found that a mosaicism threshold of 60% significantly improves accuracy, reducing false positives and negatives, and identifies niPGT-A as more representative of the actual ploidy state of embryos compared to traditional PGT-A, potentially offering the improved accuracy sought by the field.

Both PGT-A and niPGT-A depend on the successful WGA from minimal embryonic DNA. While various methodologies exist, no optimal commercial WGA kit has been determined across all applications, emphasizing the need for careful selection based on specific experimental requirements [34]. One recent study assessed NICSInst and Veriseq kits and reported similar performance in the informativeness rate and ploidy concordance rate between niPGT-A and PGT-A [28]. Our evaluation identified PicoPLEX Gold as providing enhanced library quality and ploidy accuracy compared to PG-Seq and NICSInst and supporting the establishment of a robust

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TABLE 3 | Clinical outcome of frozen–thawed embryo transfer (FET) of blastocysts diagnosed as euploid by TE biopsy.

Parameter	Euploid-TE/euploid-SCM	Euploid-TE/aneuploid-SCM	*p*
Cycles	16	12	
Patients	14	10	
SET/DET	13/3	10/2	
Transferred euploid blastocysts	19	14	
Biochemical pregnancies	14 (73.7%)	7 (50.0%)	0.162
Clinical pregnancies	11 (57.9%)	5 (35.7%)	0.207
Miscarriages[a]	0	1 (20.0%)	0.125
On-going pregnancy[b]	11 (57.9%)	4 (28.6%)	0.094
Deliveries	10	4	0.126
Singleton/twins	9/1	4/0	0.511
Live birth rate[c]	57.9%	28.6%	0.094

Abbreviations: DET, double embryo transfer; SCM, spent culture medium; SET, single embryo transfer; TE, trophectoderm.
[a]Miscarriage was defined as the spontaneous loss of a pregnancy before the 20th week.
[b]Ongoing pregnancy was defined when the pregnancy had completed ≥ 20 weeks of gestation.
[c]Live birth rate was defined as alive newborns per transferred blastocyst.

niPGT-A workflow, improving clinical management in reproductive medicine practices.

The embryonic cell-free DNA primarily stems from, dominantly but not exclusively, apoptosis [12], as well as from the active release of DNA by these developing cells [35]. Occasionally, the embryonic cfDNA can also be influenced by necrosis of embryonic cells due to oxidative stress, endotoxin, and ATP depletions [36], and maternal cumulus cells. Such a safeguard process is typically implemented during the morula and blastocyst stages of a developing embryo to maintain correct differentiation or to exclude chromosomal and nuclear anomalies [12, 21]. Although studies indicate that approximately 80% of cleavage-stage (i.e., E3) human embryos have chromosomal abnormalities [21, 37], recent research has revealed that human embryos can self-correct by eliminating aneuploid cells through BMP-4-induced apoptosis [21, 33], in a p53-dependent manner [38]. Growing evidence from various experimental models [20, 21] and real-world data [39] supports the theory of self-correction of mosaicism by differential proliferation, whereby aneuploid cells are gradually diluted to clinical irrelevance. These findings support the notion that mosaicism can be resolved during development; thus, establishing an accurate mosaicism threshold is crucial to distinguish true aneuploid embryos from misidentified euploid ones. By implementing a mosaicism threshold of 60%, the study significantly increased the specificity and accuracy of the niPGT-A, corroborating earlier research that highlighted the importance of accurate threshold settings [14, 21] to ensure precise embryo assessments and improve clinical outcomes.

Our optimized protocol achieved concordance rates of 77.9% and 76.5% between SCM and TE biopsies from frozen and fresh embryos, respectively, aligning with the 78.2% concordance rate reported in the largest cohort study of 1301 embryos [24]. Discrepancies in ploidy results can occur due to differences in embryonic development and sampling methods. Studies indicate that the aneuploid rate of embryos drops below 11% after E4, with aneuploid cells commonly found in the trophoblast lineage [21, 40]. This suggests aneuploid cells may be directed toward the trophoblast rather than the ICM [40, 41]. In our study, 52.9% of discrepancies showed a euploid state in the WB and SCM but an aneuploid state in TE cells, supporting the theory that TE cells retain aneuploid cells, while the SCM reflects the euploid state of ICM. On the other hand, discordant embryos (euploid TE/aneuploid SCM) may arise from factors such as undetected TE cell aneuploidy or mosaicism, or maternal cell contamination. Under these discordant conditions, it may be prudent to exercise caution when prioritizing these embryos for transfer. As PGT-A relies on random TE biopsies, it is susceptible to false calls, especially with low-level mosaicism [40]. In contrast, niPGT-A provides a more accurate depiction of blastocyst ploidy, showcasing its significant clinical potential.

Our study revealed that using AH significantly boosts the interpretable rate of niPGT-A findings. The AH procedure has been previously asserted to have benefits for patients with poor prognoses [42–44], and is widely employed in niPGT-A protocols (review see Cinnioglu et al.) [45]. However, previous studies also demonstrated that AH did not increase cfDNA yield on day 5 SCM [26], and SCM collected from blastocysts without AH still yielded interpretable NGS results [46]. We reasoned that the discrepancy could be at the timing of conducting the AH procedure, as the two studies performed AH on day 3 (~8-cell stage), and, in contrast, our laboratory conducted medium changeover at the 8-cell stage, performed AH at the morula stage, and collected SCM at the blastocyst stage. Mechanistically, the increased availability of cfDNA associated with AH enhances the library concentration in niPGT-A. This suggests that AH facilitates the release of cfDNA into SCM, improving the interpretability of genomic data. Additionally, our findings support the theory of embryo self-correction, where mosaicism resolves as aneuploid cells are diluted through differential proliferation during development.

In addition to the timing for conducting AH, the timing of SCM collection represents another critical, yet not fully standardized, issue that remains a point of debate. One study compared SCM collected on Day 5 or 6 after retrieval and has shown a higher concordance rate when SCM was collected on Day 6 [47]. Furthermore, the EMBRACE service collects SCM on days 6–7 after reaching the expanded blastocyst stage [25]. Although longer culture periods boost library concentrations, they may compromise embryo viability and developmental potential [48], thus outweighing the benefits associated with extended culture periods. Our niPGT-A pipeline isolates cfDNA from a 24–72 h (majority 48 h) incubation period after the D3 medium change-over,

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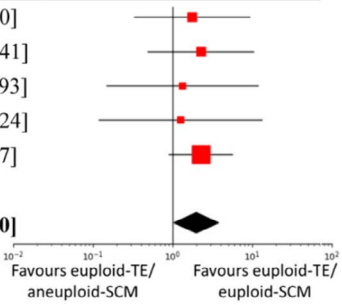
a

Clinical pregnancy

Study or subgroup	euploid-TE/ euploid-SCM Event	euploid-TE/ euploid-SCM Total	euploid-TE/ aneuploid-SCM Event	euploid-TE/ aneuploid-SCM Total	Weight (%)	Risk Ratios Fixed, 95% CI
Current study (PicoPLEX)	7	13	4	10	15.2	1.75 [0.33, 9.30]
Rubio et al. 2019 (Ion ReproSeq)	9	17	4	12	18.1	2.25 [0.49, 10.41]
Lledo et al. 2020 (Veriseq)	4	7	3	6	8.9	1.33 [0.15, 11.93]
Lledo et al. 2020 (NICS)	5	9	2	4	7.6	1.25 [0.12, 13.24]
Sun et al. 2023 (NICS)	38	52	18	33	50.2	2.26 [0.90, 5.67]
Total (95% CI)		**98**		**65**	**100**	**1.78 [1.03, 3.80]**
Total event	63		31			

Heterogeneity: Chi^2=0.40, df=4 (P=0.983); I^2=0%

Test for overall effect: Z=2.06 (P=0.039)



b

On-going pregnancy

Study or subgroup	euploid-TE/ euploid-SCM Event	euploid-TE/ euploid-SCM Total	euploid-TE/ aneuploid-SCM Event	euploid-TE/ aneuploid-SCM Total	Weight (%)	Risk Ratios Fixed, 95% CI
Current study (PicoPLEX)	7	13	3	10	14.7	2.72 [0.48, 15.47]
Rubio et al. 2019 (Ion ReproSeq)	9	17	2	12	13.9	5.63 [0.94, 33.77]
Lledo et al. 2020 (Veriseq)	4	7	2	6	8.7	2.67 [0.28, 25.64]
Lledo et al. 2020 (NICS)	4	9	2	4	8.0	0.80 [0.08, 8.47]
Sun et al. 2023 (NICS)	36	52	17	33	54.7	2.12 [0.86, 5.22]
Total (95% CI)		**98**		**65**	**100**	**2.37 [1.22, 4.63]**
Total event	60		26			

Heterogeneity: Chi^2=1.80, df=4 (P=0.772); I^2=0%

Test for overall effect: Z=2.54 (P=0.011)

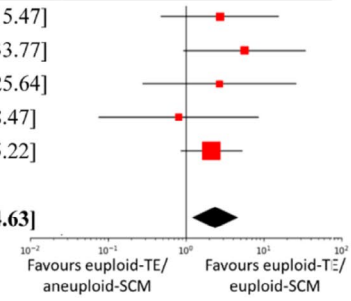


FIGURE 5 | Meta-analysis of pregnancy outcomes guided by embryo ploidy status. Forest plots from meta-analysis studies assessing pregnancy outcomes for transfers guided by euploid-TE/euploid-SCM and euploid-TE/aneuploid-SCM: Clinical pregnancy (A), and Ongoing pregnancy (B). A fixed-effects model was applied, as there was no statistically significant heterogeneity (p of $\chi^2 > 0.05$; $I^2 = 0\%$). Results are presented as risk ratios with 95% confidence intervals. The p value for the overall effect size was calculated using a two-tailed z-test. The individual studies are listed for all plots, and their contribution to the overall result is weighted accordingly.

offering flexibility within standard two-step embryo culture systems. Although the balance between library concentration and embryo viability remains an ongoing discussion, our dynamic timing, developmental stage-based workflow enables seamless integration into clinical practice, making it suitable for routine clinical application.

Previous studies showed live birth rates for euploid niPGT-A transfers at 55.6% [16], 52.0% [49], and 43.9% [50]. Higher ongoing implantation rates were observed for euploid-TE/euploid-SCM compared to euploid-TE/aneuploid-SCM, at 52.9% [25] and 57.1% [28], respectively. Our study, which included 16 and 12 FET cycles using these configurations, yielded live birth rates of 57.9% and 28.6%. Despite limited statistical significance due to small sample sizes, a meta-analysis combining our data with three previous studies showed a significantly elevated clinical and ongoing pregnancy rate for euploid-TE/aneuploidy-SCM transfers. These findings underscore the potential of niPGT-A to enhance decision-making for embryo transfers compared to PGT-A, without the associated risks of embryo biopsy.

Our findings align with recent studies indicating the potential of niPGT-A to provide accurate chromosomal assessments without the need for invasive biopsies. However, our study uniquely demonstrates the efficacy of a 60% mosaicism threshold in improving specificity and overall accuracy, surpassing traditional PGT-A methods. This may lead to more informed decisions regarding embryo transfer and improve clinical outcomes, with potential benefits noted in higher live birth rates from euploid-SCM-guided transfers. Furthermore, integrating AH into the protocol is advised to enhance the effectiveness of niPGT-A, particularly for patients with challenging prognoses.

While promising, our study is inherently limited by its small sample size and single-center design, which may restrict the generalizability of our findings. Additionally, variability in patient

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demographics and clinical practices can impact procedural consistency across different settings. Furthermore, the challenge of determining the optimal timing for SCM collection and its impact on embryo viability, as well as the long-term developmental outcomes of embryos selected via niPGT-A, remains a critical area requiring further investigation. Of note, while our study demonstrates a high sex concordance rate between TE biopsy and SCM samples, the X:Y chromosome ratio-based estimation of maternal cell contamination is limited to XY embryos. As a result, contamination levels in XX embryos could not be directly assessed using this method. Future studies should prioritize larger, multi-center trials with diverse patient populations to confirm these findings and assess the widespread applicability of the niPGT-A protocol. Moreover, research should focus on refining mosaicism thresholds based on specific chromosomal abnormalities and evaluating the correlation between cfDFI and embryo developmental potential. Finally, alternative strategies, such as utilizing cell-type-specific methylation patterns to distinguish maternal and embryonic DNA [51], should be studied and employed to quantify maternal cell contamination in both XX and XY embryos. Nonetheless, comparative studies assessing the cost-effectiveness of niPGT-A versus traditional PGT-A would be valuable for informing clinical adoption and resource allocation. Addressing these limitations and pursuing these future directions will be crucial for realizing the full potential of niPGT-A in routine IVF practice and providing patients with a safer and more efficacious option for genetic screening.

In conclusion, we developed a highly accurate niPGT-A pipeline for clinical use, focusing on critical factors impacting assay performance. Our clinically validated protocol demonstrates promising accuracy and clinical outcomes through workflow optimization that includes data processing and interpretation. As niPGT-A contributes to safer IVF practices and minimizes the risk of embryo damage, potentially improving pregnancy outcomes and expanding access to PGT-A for more patients, the optimized niPGT-A procedure is recommended for implementation in routine clinical practice to enhance clinical management in future reproductive medicine.

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Ethics Statement

All procedures followed were in accordance with the ethical standards of the responsible committee on human experimentation (institutional and national) and with the Helsinki Declaration of 1964 and its later amendments. This study was approved by the Institutional Review Boards (IRB, A-ER-110-137) from the National Cheng Kung University Hospital, Taiwan.

Consent

Written informed consent was obtained from all participants for being included in the study, and the study adhered to relevant guidelines and regulations.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author, Sunny Sun, upon reasonable request.

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